

COLD BAY, AK, USAWMO: **703160**Lat: **55.221N**Lon: **162.732W**Elev: **78**StdP: **14.65**Time Zone: **-9.00 (NAK)**Period: **94-19**WBAN: **25624****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	12.4	3.7	6.7	9.9	7.0	7.9	13.3	46.1	37.2	41.0	34.4	15.7	340	0.955

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	61.0	55.7	58.6	54.3	56.8	53.2	56.9	60.0	55.3	57.5	54.0	55.9	14.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
55.3	65.3	57.7	54.1	62.4	56.4	52.8	59.6	55.2	24.3	60.0	23.4	57.6	22.6	55.9	67.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
39.9	35.0	32.0	DB	3.9	66.8	6.1	3.5	-0.5	69.3	-4.0	71.4	-7.5	73.4	-11.9	75.9
			WB	3.1	60.2	5.9	2.8	-1.2	62.1	-4.6	63.7	-8.0	65.3	-12.3	67.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	28.8	30.7	29.6	34.8	40.7	46.8	51.3	52.5	48.4	41.3	35.3	30.9	(6)
	DBStd	10.10	8.00	7.26	7.68	5.35	4.32	3.25	2.96	3.22	3.55	4.86	5.84	6.96	(7)
	HDD50	4081	659	541	632	457	288	103	19	70	270	441	591		(8)
	HDD65	9377	1124	961	1097	907	752	546	425	386	498	734	891	1056	(9)
	CDD50	180	0	0	0	0	1	8	59	89	22	1	0	0	(10)
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
	WSAvg	15.8	15.8	17.1	16.2	16.6	15.5	14.7	14.3	14.5	15.5	15.9	17.0	16.8	(14)
	PrecAvg	39.1	3.1	2.6	2.5	2.4	2.4	2.5	3.5	4.4	4.6	4.6	4.1		(15)
(16) Precipitation	PrecMax	84.3	6.3	7.6	7.1	6.5	6.9	8.3	6.1	6.5	9.8	10.2	12.2	13.9	(16)
	PrecMin	23.4	0.6	0.4	0.0	0.2	0.3	0.1	0.4	1.1	1.3	1.5	1.1	1.0	(17)
	PrecStd	9.6	1.4	1.5	1.5	1.5	1.3	1.5	1.2	1.3	1.7	1.8	2.2	2.2	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.1	43.8	44.5	48.4	58.0	60.1	64.5	65.3	59.2	54.2	50.4	46.9		(19)
	0.4% MCWB	40.4	41.0	40.4	42.4	48.3	52.3	57.1	58.6	54.8	50.0	47.3	44.3		(20)
	2% DB	41.2	41.6	42.1	45.6	51.9	56.4	60.3	61.7	56.8	51.9	47.7	43.6		(21)
	2% MCWB	39.4	39.6	39.2	41.0	45.4	50.2	54.9	57.1	53.3	49.1	45.7	41.6		(22)
	5% DB	39.6	40.0	40.3	43.6	49.1	54.0	58.0	59.2	55.1	50.1	45.7	41.5		(23)
	5% MCWB	37.9	38.3	37.8	39.8	44.0	48.8	53.7	55.5	52.2	47.5	43.6	39.8		(24)
	10% DB	38.1	38.6	38.7	41.8	46.9	52.0	55.9	57.2	53.7	48.5	43.6	39.8		(25)
	10% MCWB	36.6	37.0	36.5	38.9	43.0	47.7	52.6	54.2	51.0	45.9	41.7	38.1		(26)
	0.4% WB	41.4	42.1	42.1	44.1	49.1	53.6	58.4	60.0	57.1	51.5	48.3	45.0		(27)
	0.4% MCDB	42.3	43.0	43.2	46.8	55.8	58.4	63.0	63.3	58.2	52.6	49.7	46.4		(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	39.7	40.2	39.9	42.0	46.4	51.2	55.7	57.6	54.4	50.0	46.2	42.2		(29)
	2% MCDB	40.8	41.0	41.4	44.5	50.8	54.5	58.8	61.0	55.6	51.5	47.6	43.4		(30)
	5% WB	38.2	38.6	38.2	40.5	45.0	49.7	54.3	56.1	52.9	48.0	44.1	40.2		(31)
	5% MCDB	39.3	39.6	39.7	42.9	48.2	52.8	56.9	58.6	54.3	49.8	45.5	41.4		(32)
	10% WB	36.9	37.2	36.9	39.2	43.7	48.3	53.1	54.8	51.4	46.3	42.0	38.3		(33)
	10% MCDB	38.0	38.3	38.4	41.4	46.4	51.3	55.3	56.7	53.0	48.3	43.5	39.6		(34)
(36) Mean Daily Temperature Range	MDBR	8.1	8.3	9.7	9.1	9.7	8.9	8.1	8.1	8.5	8.9	7.9	8.2		(35)
	5% DB MCDBR	7.0	8.1	9.7	11.3	13.6	12.8	11.8	11.0	10.0	9.8	8.7	7.7		(36)
	5% DB MCWBR	6.3	7.1	7.8	8.1	8.8	7.8	7.4	7.4	7.3	8.1	8.1	7.5		(37)
	5% WB MCDBR	6.6	7.6	8.8	10.0	11.9	11.3	10.6	10.1	8.5	8.9	8.5	7.3		(38)
	5% WB MCWBR	6.3	7.2	7.6	7.7	8.2	7.3	7.2	7.2	7.1	8.1	8.4	7.5		(39)
(40) Clear-Sky Solar Irradiance	taub	0.299	0.336	0.343	0.387	0.402	0.384	0.383	0.386	0.368	0.347	0.324	0.308		(40)
	taud	2.251	2.245	2.275	2.197	2.219	2.375	2.432	2.413	2.382	2.305	2.186	2.137		(41)
	Ebn at Noon	197	230	261	263	266	273	272	263	251	223	178	155		(42)
	Edh at Noon	18	26	32	40	41	36	33	32	29	24	19	16		(43)
(44) All-Sky Solar Radiation	RadAvg	154	331	756	1105	1365	1419	1241	1040	781	453	194	110		(44)
	RadStd	18	37	108	85	121	137	142	94	66	25	19	11		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	+2.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)			N/A	+2.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page